Cyber Security

Networking Fundamentals

for Cyber Security

Exam-Oriented Study Notes

BCA, MAKAUT

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1 Introduction to Networking

Definition

A **computer network** is a collection of interconnected devices (computers, servers, phones, printers, routers) that share data, resources and services over wired or wireless links using agreed rules called **protocols**.

1.1 Goals and Components

- **Goals:** resource sharing, communication, reliability, scalability, cost saving.
- **Components:** nodes (hosts), links (media), networking devices, protocols, and network software/OS.

1.2 Types of Networks (by Size)

Type	Coverage	Example / Note
PAN	Few metres	Bluetooth between phone and earbuds
LAN	Room, building, campus	College computer lab, home Wi-Fi; high speed, privately owned
MAN	City	Cable TV network, city-wide ISP backbone
WAN	Country / world	The Internet; uses leased lines, satellite, fibre

1.3 Network Topologies

Topology	Layout	Advantage	Disadvantage
Bus	Single shared cable	Cheap, easy to install	One cable fault stops all; collisions
Star	All nodes connect to a central switch	Easy to manage; one node failure is isolated	Central device is single point of failure
Ring	Each node connects to two neighbours	Orderly access (token)	One break can disrupt the ring
Mesh	Every node linked to many others	Very reliable; multiple paths	Costly; complex cabling
Tree / Hybrid	Combination of topologies	Scalable; hierarchical	Complex to maintain

1.4 Network Models

- **Client–Server:** central server provides services; clients request them (web, email). Easier to secure and manage centrally.
- **Peer-to-Peer (P2P):** every node acts as both client and server. Cheap but harder to control and secure.

Security Angle

Every connected device is a possible **attack surface**. Networking knowledge is needed to understand how attacks such as sniffing, spoofing, scanning and denial of service work, and how the **CIA triad** (Confidentiality, Integrity, Availability) is protected in transit.

Exam Tip

Draw a neat diagram for each topology when asked (3–5 marks). Always add one advantage and one disadvantage.

Practice Questions

Short Answer (2 marks)

1. Define a computer network.
2. Differentiate between LAN and WAN.
3. Why is a star topology preferred in modern offices?

Long Answer (5–8 marks)

4. Explain the different types of network topologies with diagrams, advantages and disadvantages.
5. Compare client–server and peer-to-peer models. Which is more secure and why?

2 OSI and TCP/IP Framework

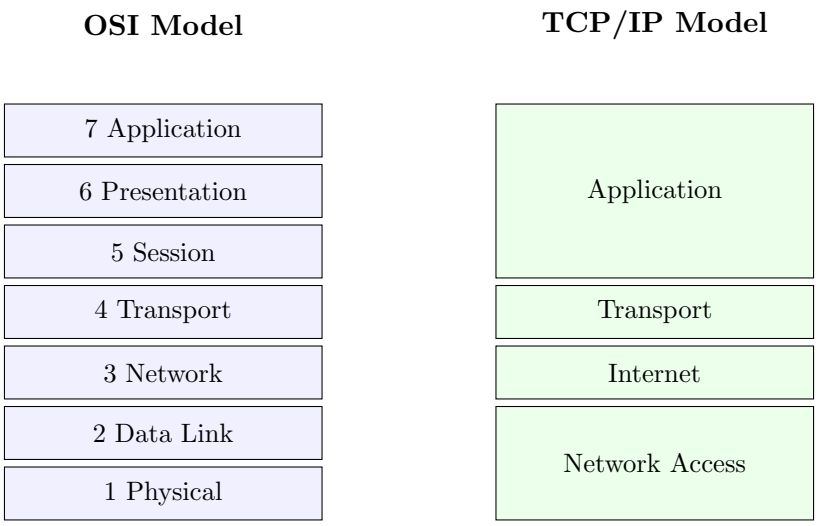
2.1 OSI Reference Model

The **Open Systems Interconnection (OSI)** model, developed by ISO, divides network communication into **7 layers**. Each layer serves the layer above and uses the layer below.
Mnemonic (top to bottom): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

No.	Layer	PDU	Main Function	Examples
7	Application	Data	User interface to network services	HTTP, SMTP, DNS, FTP
6	Presentation	Data	Translation, encryption, compression	SSL/TLS, JPEG, ASCII
5	Session	Data	Establish, manage, terminate sessions	NetBIOS, RPC
4	Transport	Segment / Datagram	End-to-end delivery, reliability, ports	TCP, UDP
3	Network	Packet	Logical addressing and routing	IP, ICMP, routers
2	Data Link	Frame	Physical addressing (MAC), error detection	Ethernet, switches, ARP
1	Physical	Bits	Transmission of raw bits over medium	Cables, hubs, Wi-Fi radio

2.2 TCP/IP Model

The **TCP/IP model** is the practical protocol suite of the Internet and has **4 layers**: Application, Transport, Internet, and Network Access (Link).



2.3 OSI vs TCP/IP

Basis	OSI	TCP/IP
Layers	7	4
Nature	Theoretical reference model	Practical, protocol-based model
Development	ISO; model first, protocols later	DARPA; protocols first, model later
Session / Presentation	Separate layers	Merged into Application layer
Transport service	Connection-oriented only (in theory)	TCP (connection-oriented) and UDP (connectionless)
Usage	Teaching, troubleshooting	Real-world Internet

Security Angle

Attacks are classified by layer. **L1:** cable tapping. **L2:** MAC flooding, ARP spoofing. **L3:** IP spoofing, ICMP flood. **L4:** SYN flood, port scanning. **L7:** phishing, SQL injection, DNS poisoning.

Exam Tip

"Compare OSI and TCP/IP" is a very frequent 5-mark question. Write at least 5 differences in table form and draw the layer-mapping diagram.

Practice Questions

Short Answer: (1) What is a PDU? Name the PDU at each OSI layer. (2) Which OSI layers merge into the TCP/IP Application layer? (3) Name one attack at the Data Link layer.

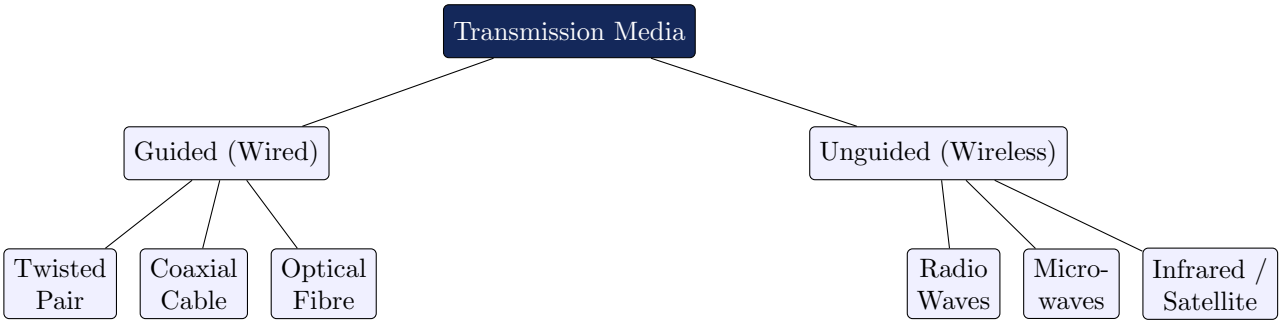
Long Answer: (4) Explain the seven layers of OSI with functions and examples. (5) Compare OSI and TCP/IP with a neat diagram.

3 Transmission Media

Definition

Transmission media is the physical path (or wireless channel) through which data travels from sender to receiver. It operates at the **Physical layer**.

3.1 Classification



3.2 Guided Media

- **Twisted Pair:** pairs of copper wires twisted to reduce interference. **UTP** (unshielded) is cheap and common in LANs; **STP** (shielded) resists noise better. Max segment length is **100 m**.
- **Coaxial Cable:** central copper conductor, insulator, metal shield, outer jacket. Used in cable TV and older Ethernet; better shielding than UTP.
- **Optical Fibre:** data sent as light pulses through glass/plastic core. **Single-mode (SMF)** for long distance; **multi-mode (MMF)** for shorter distance inside buildings or campuses.

3.3 Unguided Media

- **Radio waves:** omnidirectional, pass through walls; Wi-Fi, Bluetooth, FM radio.
- **Microwaves:** line-of-sight, point-to-point links, mobile backhaul.
- **Infrared:** very short range, cannot pass walls; remote controls.
- **Satellite:** wide coverage, high latency; GPS, VSAT links.

3.4 Comparison of Guided Media

Feature	UTP	Coaxial	Optical Fibre
Signal	Electrical	Electrical	Light
Speed / bandwidth	Up to 10 Gbps (Cat6a)	Moderate	Very high (Tbps possible)
Distance	100 m per segment	Hundreds of metres	Kilometres
EMI immunity	Low	Medium	Immune
Cost	Low	Medium	High
Tapping difficulty	Easy	Moderate	Hard (needs cutting or bending the fibre)

Security Angle

Copper cables radiate signals that can be **tapped**; fibre is the most secure wired medium. Wireless media can be **eavesdropped, jammed** or attacked with rogue access points, so strong encryption (WPA2/WPA3) is essential.

Case Study: Campus Network Cabling

Scenario: A college has three buildings: Admin, Science Block (labs) and Hostel, about 300 m apart, plus 120 lab PCs.

Design decisions:

- **Inside labs:** Cat6 UTP from PCs to the floor switch (cheap, within 100 m).
- **Between buildings:** single-mode fibre backbone (300 m is beyond UTP's 100 m limit, and fibre is immune to lightning and EMI between buildings).
- **Hostel:** Wi-Fi access points with WPA3 for flexibility.

Result: a cost-effective hybrid network with a secure, high-speed backbone. **Lesson:** choose media by distance, bandwidth, cost and security needs.

Exam Tip

For case-study questions, structure the answer as *Scenario* → *Problem* → *Medium chosen* → *Justification* → *Conclusion*.

Practice Questions

Short Answer: (1) Why are cables twisted in twisted-pair? (2) Differentiate SMF and MMF. (3) Why is fibre considered more secure?

Long Answer: (4) Explain guided and unguided media with examples. (5) Recommend and justify transmission media for a multi-building campus.

4 Network Devices

Device	Layer	Function	Key Point
Repeater	1	Regenerates and amplifies signal	Extends distance; no filtering
Hub	1	Broadcasts incoming data to all ports	One collision domain; insecure
Bridge	2	Connects two LAN segments, filters by MAC	Fewer collisions
Switch	2 (3)	Forwards frames to the correct port using MAC table	Each port is its own collision domain
Router	3	Forwards packets between networks using IP	Separates broadcast domains; uses routing table
Gateway	4–7	Connects networks with different protocols	Protocol translation
Modem	1	Converts digital to analog and back	Connects to ISP
Access Point	2	Connects wireless clients to wired LAN	Needs encryption
Firewall	3–7	Filters traffic by security rules	First line of defence
IDS / IPS	3–7	Detects (IDS) or blocks (IPS) malicious activity	Monitoring and prevention

4.1 Collision vs Broadcast Domain

- **Hub:** all ports share one collision domain and one broadcast domain.
- **Switch:** each port = separate collision domain; all ports = one broadcast domain (without VLANs).
- **Router:** each interface = separate broadcast domain.

4.2 Firewall Types

Packet filtering (checks IP/port), **stateful inspection** (tracks connections), **proxy / application** firewall (inspects content), and **Next-Generation Firewall (NGFW)** (adds deep inspection, IPS, application awareness).

Security Angle

Hubs allow easy **sniffing**, since every device sees all traffic. Switches reduce this but can still be attacked by **MAC flooding** and **ARP spoofing**. Port security, VLANs and DHCP snooping are common defences.

Case Study: Securing a Small Office Network

Scenario: A 30-employee office uses one flat network with a hub; a guest laptop was found capturing traffic.
Redesign:

1. Replace the hub with a **managed switch**; enable port security to limit MAC addresses per port.
2. Create **VLANs**: Staff, Servers, Guest.
3. Place a **router with a firewall** at the Internet edge; allow only required ports.
4. Add a **WPA3 access point** with a separate guest SSID.
5. Deploy an **IDS/IPS** to monitor suspicious traffic.

Result: sniffing is no longer practical, guests are isolated, and the Internet edge is controlled.

Exam Tip

"Difference between hub, switch and router" appears often. Compare on layer, addressing used, collision/broadcast domains, intelligence and security.

Practice Questions

Short Answer: (1) Why is a switch better than a hub? (2) What is the role of a firewall? (3) Which device separates broadcast domains?
Long Answer: (4) Explain the functions of repeater, hub, switch, router and gateway. (5) Explain firewall types and their placement in a network.

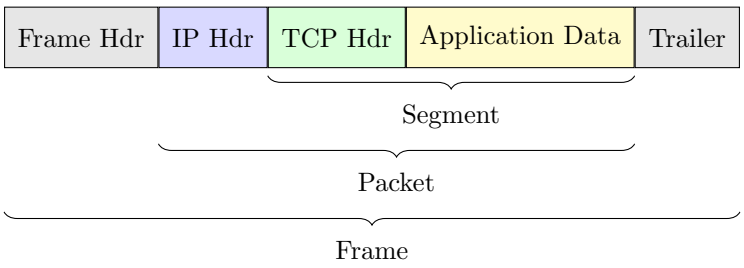
5 TCP/IP Framework in Detail

5.1 Layer-wise Overview

Layer	Role	Protocols
Application	Services for user applications	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SSH, Telnet
Transport	Process-to-process delivery using ports	TCP, UDP
Internet	Logical addressing and routing	IP, ICMP, ARP, IGMP, IPsec
Network Access	Frame delivery on local link	Ethernet, Wi-Fi (802.11), PPP

5.2 Encapsulation

Data moves down the stack and each layer adds its own header (and a trailer at the link layer). The receiver removes them in reverse (**de-encapsulation**).



5.3 Transport Layer: TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Acknowledgements, retransmission	No guarantee
Ordering	Sequence numbers keep order	No ordering
Flow / congestion control	Yes (window)	No
Header size	20–60 bytes	8 bytes
Speed	Slower	Faster
Uses	Web, email, file transfer, SSH	DNS, streaming, VoIP, gaming, DHCP

TCP header fields: source port, destination port, sequence number, acknowledgement number, header length, **flags** (URG, ACK, PSH, RST, SYN, FIN), window size, checksum, urgent pointer, options.

Port ranges: well-known **0–1023**, registered **1024–49151**, dynamic/private **49152–65535**.

5.4 Internet Layer

- **IP (Internet Protocol):** connectionless, unreliable, best-effort delivery with logical (IP) addressing.
- **ICMP:** error reporting and diagnostics (`ping`, `traceroute`).
- **ARP:** maps an IP address to a MAC address on the local network.
- **IGMP:** manages multicast group membership.

IPv4 Address Classes

Class	First Octet	Default Mask	Prefix	Use
A	1–126	255.0.0.0	/8	Very large networks
B	128–191	255.255.0.0	/16	Medium networks
C	192–223	255.255.255.0	/24	Small networks
D	224–239	–	–	Multicast
E	240–255	–	–	Experimental

Private ranges (RFC 1918): `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`. **Loopback:** `127.0.0.1`.

IPv4 vs IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal	Hexadecimal with colons
Address space	About 4.3×10^9	About 3.4×10^{38}
Header	Variable, with checksum	Fixed 40 bytes, no checksum
Security	IPsec optional	IPsec built into the design
Configuration	Manual / DHCP	Auto-configuration (SLAAC)

5.5 Network Access Layer

Handles **MAC addressing** (48-bit hardware address), framing, and error detection (CRC). Technologies: Ethernet (802.3), Wi-Fi (802.11).

Security Angle

ARP spoofing (fake ARP replies) enables man-in-the-middle attacks; **IP spoofing** forges source addresses; **ICMP flood / ping of death** cause denial of service. **IPsec** secures traffic at the Internet layer.

Exam Tip

Learn the TCP vs UDP table and the encapsulation diagram well. Port numbers of common services are also asked as 1-mark questions (see Section 9).

Practice Questions

Short Answer: (1) What is encapsulation? (2) Why does DNS mainly use UDP? (3) What does ARP do?

Long Answer: (4) Explain the TCP/IP model layer by layer. (5) Compare TCP and UDP and IPv4 and IPv6.

6 Subnet Masking

Definitions

Subnetting divides one large network into smaller sub-networks (subnets).
A **subnet mask** is a 32-bit value that separates the **network portion** from the **host portion** of an IP address. Network bits are 1s and host bits are 0s.
CIDR notation writes the number of network bits after a slash, e.g. `192.168.1.0/24`.

6.1 Why Subnet?

Efficient use of IP addresses, smaller broadcast domains (better performance), easier management, and **security isolation** between departments.

6.2 Key Formulas

$$\text{Number of subnets} = 2^n \quad (n = \text{bits borrowed from host part})$$
$$\text{Hosts per subnet} = 2^h - 2 \quad (h = \text{remaining host bits})$$
$$\text{Block size} = 256 - (\text{interesting octet of mask})$$

The "-2" removes the **network address** (all host bits 0) and the **broadcast address** (all host bits 1).

6.3 Common CIDR Values

CIDR	Subnet Mask	Total Addresses	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

6.4 Worked Example 1: Find Network, Broadcast and Host Range

Given: IP address `192.168.1.130` with mask `255.255.255.192` (/26).

Step 1: Bitwise AND of IP and mask (last octet).

IP (130)	`10000010`
Mask (192)	`11000000`
AND result	`10000000` = 128

Step 2: Block size = 256 − 192 = 64. Subnets start at 0, 64, 128, 192. The IP 130 lies in the block starting at **128**.

Network address	`192.168.1.128`
First usable host	`192.168.1.129`
Last usable host	`192.168.1.190`
Broadcast address	`192.168.1.191`
Usable hosts	$2^6 - 2 = 62$

6.5 Worked Example 2: Divide a Network into 4 Subnets

Given: `192.168.10.0/24`; divide into 4 equal subnets.

Solution: $2^n \geq 4 \Rightarrow n = 2$ bits borrowed. New prefix = 24 + 2 = **/26**, mask `255.255.255.192`. Host bits = 6, so hosts per subnet = $2^6 - 2 = 62$.

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Case Study: Subnetting a Company Network

Scenario: A company has the block `192.168.20.0/24` and four departments: HR (20 hosts), Finance (25 hosts), IT (50 hosts) and Guest Wi-Fi (10 hosts).
Analysis: The largest need is 50 hosts, so $2^6 - 2 = 62$ is enough; one /26 per department works (4 equal subnets).

Department	Subnet	Usable Range	Broadcast
IT	192.168.20.0/26	.1 – .62	.63
Finance	192.168.20.64/26	.65 – .126	.127
HR	192.168.20.128/26	.129 – .190	.191
Guest	192.168.20.192/26	.193 – .254	.255

Security benefit: Firewall or ACL rules can now block Guest-to-Finance traffic, and a broadcast storm in one department does not affect the others.
Improvement: With VLSM, Guest could use a /28 and save addresses for growth.

Exam Tip

Always show: *formula* → *borrowed bits* → *new mask* → *block size* → *table of ranges*. Numericals carry 5–8 marks and step-wise marking is common. Do not forget to subtract 2 for hosts.

Practice Questions

Short Answer: (1) What is a subnet mask? (2) Why subtract 2 from host count? (3) Write `255.255.255.224` in CIDR form.

Numerical: (4) For `172.16.5.77/27`, find network, broadcast, and usable host range. (5) Divide `10.0.0.0/24` into 8 subnets.

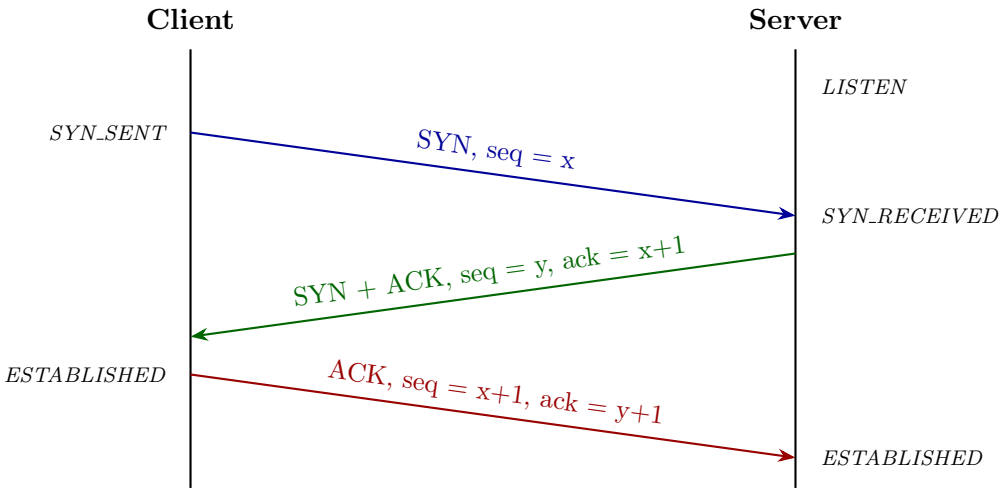
Long Answer: (6) Explain subnetting with advantages and a worked example.

7 Three-Way Handshake

Definition

The **three-way handshake** is the process TCP uses to establish a reliable connection between client and server by exchanging **SYN, SYN-ACK and ACK** segments. It synchronises sequence numbers on both sides before data transfer.

7.1 Steps



1. **SYN:** the client sends a segment with the SYN flag and its initial sequence number x.
2. **SYN-ACK:** the server replies with SYN and ACK flags, its own initial sequence number y, and acknowledgement $x + 1$.
3. **ACK:** the client acknowledges with ack $= y + 1$. The connection is now **ESTABLISHED** and data can flow.

7.2 Why Three Steps?

Both sides must confirm that they can **send and receive** and agree on **initial sequence numbers**. Two steps cannot confirm the client's ability to receive, and old duplicate SYNs could create false connections.

7.3 Connection Termination (Four-Way)

FIN $\rightarrow$ **ACK** $\rightarrow$ **FIN** $\rightarrow$ **ACK**: each direction is closed separately. **RST** abruptly resets a connection.

7.4 TCP Flags Quick Reference

Flag	Meaning
SYN	Start / synchronise a connection
ACK	Acknowledge received data
FIN	Graceful close
RST	Abort / reset connection
PSH	Push data to application immediately
URG	Urgent data

Security Angle

SYN flood (DoS): the attacker sends many SYNs with spoofed IPs and never completes the third step. The server's half-open connection queue fills up and legitimate users are refused. **Defences:** SYN cookies, shorter timeouts, rate limiting, and firewalls/IPS. **Nmap** uses SYN ("half-open") scans for port discovery: a SYN-ACK reply means the port is open, an RST means closed.

Exam Tip

Always draw the sequence diagram with seq/ack numbers (the "$x + 1$" and "$y + 1$" earn marks). Mention one security issue (SYN flood) for full marks.

Practice Questions

Short Answer: (1) Name the three segments in a TCP handshake. (2) What is a half-open connection? (3) Which flag is used to abort a connection?

Long Answer: (4) Explain the TCP three-way handshake with a diagram. (5) What is a SYN flood attack? How can it be prevented?

8 Network Address Translation (NAT)

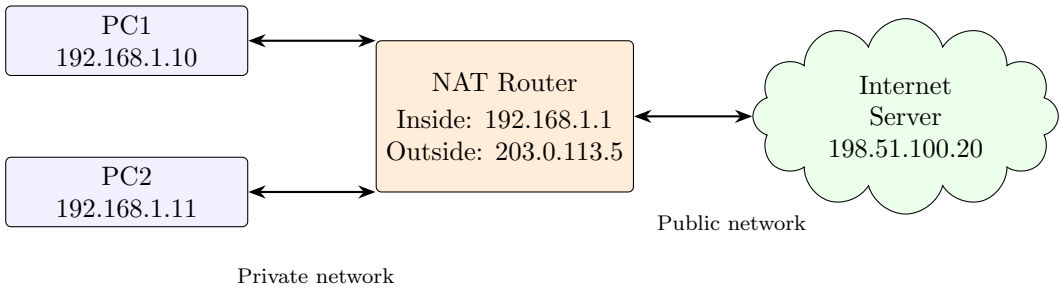
Definition

NAT is a technique, usually performed by a router or firewall, that translates **private IP addresses** used inside a network into **public IP addresses** for communication over the Internet, and back again.

8.1 Need for NAT

IPv4 has only about 4.3 billion addresses. NAT lets many private hosts share one (or a few) public IPs, which **conserves address space**.

8.2 How NAT Works



Example NAT (PAT) translation table:

Inside Local (private)	Inside Global (public)	Destination
192.168.1.10 : 50001	203.0.113.5 : 40001	198.51.100.20 : 80
192.168.1.11 : 50002	203.0.113.5 : 40002	198.51.100.20 : 80

The router rewrites the **source IP and port** on outgoing packets and uses the table to forward replies back to the correct internal host.

8.3 Types of NAT

Type	Mapping	Use
Static NAT	One private IP ↔ one fixed public IP	Hosting an internal server publicly
Dynamic NAT	Private IPs mapped from a *pool* of public IPs	Organisations with several public IPs
PAT (NAT Overload)	Many private IPs → one public IP, distinguished by **port numbers**	Homes and most offices

8.4 Advantages and Disadvantages

Advantages	Disadvantages
Saves public IPv4 addresses	Breaks true end-to-end connectivity
Hides internal addressing from outsiders	Adds processing delay
Flexible internal addressing	Complicates VoIP, VPN, and P2P (needs NAT traversal)
Easy to change ISP without renumbering	Makes logging and tracing hosts harder

Security Angle

NAT blocks unsolicited inbound connections by default, which gives some protection, but **NAT is not a firewall**. Port forwarding, malware making outbound connections, and phishing bypass it. A proper firewall is still required. IPv6 largely removes the need for NAT.

Exam Tip

Remember: PAT = many-to-one using ports. Draw the diagram with private and public addresses and show a small translation table.

Practice Questions

Short Answer: (1) Define NAT. (2) What is PAT? (3) Is NAT a security feature or a firewall?
Long Answer: (4) Explain NAT, its types, and working with a diagram. (5) Discuss advantages and limitations of NAT.

9 Different Protocols and Their Uses

A **protocol** is a set of rules governing how devices format, send, receive and acknowledge data.

9.1 Important Protocols and Port Numbers

Protocol	Port	Transport	Use
HTTP	80	TCP	Web pages (unencrypted)
HTTPS	443	TCP	Secure web using TLS
FTP	20, 21	TCP	File transfer (plain text)
SFTP	22	TCP	Secure file transfer over SSH
SSH	22	TCP	Secure remote login
Telnet	23	TCP	Remote login (plain text, insecure)
SMTP	25 (587)	TCP	Sending email
POP3	110	TCP	Downloading email
IMAP	143	TCP	Accessing email on server
DNS	53	UDP / TCP	Domain name to IP resolution
DHCP	67, 68	UDP	Automatic IP configuration
TFTP	69	UDP	Simple file transfer
NTP	123	UDP	Time synchronisation
SNMP	161, 162	UDP	Network device monitoring
LDAP	389	TCP	Directory services
SMB	445	TCP	Windows file/printer sharing
RDP	3389	TCP	Remote desktop
Syslog	514	UDP	Log collection
ICMP	–	(Network layer)	Diagnostics and errors (ping)
ARP	–	(Link/Network)	IP to MAC resolution

9.2 Key Protocols Explained

- **DNS:** the Internet's "phonebook". Vulnerable to cache poisoning and spoofing; defended by DNSSEC.
- **DHCP:** automatically assigns IP, mask, gateway and DNS using the DORA process (**D**iscover, **O**ffer, **R**equest, **A**cknowledge). Risk: rogue DHCP servers.
- **HTTP vs HTTPS:** HTTPS wraps HTTP inside **TLS**, giving encryption, integrity and server authentication.
- **SSH:** encrypted replacement for Telnet, supporting key-based authentication and tunnelling.
- **Email:** SMTP **sends**; POP3 **downloads** (usually removes from server); IMAP **syncs** with the server.
- **ICMP:** used by `ping` and `traceroute`; also abused for flooding and reconnaissance.

9.3 Insecure vs Secure Alternatives

Insecure	Secure Replacement	Reason
HTTP	HTTPS	Encrypts web traffic
Telnet	SSH	Encrypts remote sessions
FTP	SFTP / FTPS	Protects credentials and files
POP3 / IMAP / SMTP	POP3S / IMAPS / SMTPS	Encrypts email protocols
SNMPv1/v2	SNMPv3	Adds authentication and encryption

Exam Tip

Port numbers are easy 1-mark questions. Learn the table, and be ready to state the **secure alternative** to each insecure protocol, as examiners like security-oriented answers in this subject.

Practice Questions

Short Answer: (1) Give the port numbers of HTTP, HTTPS, SSH and DNS. (2) Why is Telnet considered insecure? (3) What is DORA?

Long Answer: (4) Explain any five application layer protocols with their uses and ports. (5) Differentiate SMTP, POP3 and IMAP.

10 Quick Revision Sheet

Topic	Remember
OSI vs TCP/IP	7 vs 4 layers; OSI theoretical, TCP/IP practical
PDUs	Data, Segment, Packet, Frame, Bits
UTP limit	100 m; fibre for long distance and best security
Hub / Switch / Router	Layer 1 / Layer 2 / Layer 3
Hosts per subnet	$2^h - 2$
Block size	256− mask octet
Three-way handshake	SYN → SYN-ACK → ACK
SYN flood	Half-open connections exhaust server; use SYN cookies
NAT / PAT	Private to public; PAT uses ports (many-to-one)
Private IP ranges	10/8, 172.16/12, 192.168/16
Secure pairs	HTTP→HTTPS, Telnet→SSH, FTP→SFTP